Functions & Transformations

Section 1: Domain, Range & Function Types

1 What is the domain of the real-valued function $f(x) = \sqrt{2x - 8}$?

a. $[4, \infty)$

b. $(4, \infty)$

c. $[2, \infty)$

d. $(-\infty, 4]$

2 Which of the following functions is an ODD function?

a. $f(x) = x^3 - 5x$

b. $f(x) = x^4 + 2x^2$

c. $f(x) = x^2 - 3$

d. $f(x) = x^3 + 1$

3 What is the vertical asymptote of the rational function $f(x) = \frac{3x+1}{2x-6}$?

a. $x = 3$

b. $x = 2$

c. $x = -3$

d. $y = \frac{3}{2}$

Section 2: Composite Functions & Inverse Functions

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4 If $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, what is $(f \circ g)(3)$?

a. 19

b. 17

c. 23

d. 15

5 What is the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x - 5}{2}$?

a. $\frac{2x + 5}{3}$

b. $\frac{2x - 5}{3}$

c. $\frac{3x + 5}{2}$

d. $\frac{x + 5}{6}$

6 If $f(x) = x^3 - 2$, what is the value of $f^{-1}(25)$?

a. 3

b. 2

c. 4

d. 5

Section 3: Geometric Transformations of Graphs

7 How is the graph of $g(x) = (x - 3)^2 + 4$ obtained from the parent function $f(x) = x^2$?

a. Shifted 3 units right and 4 units up

b. Shifted 3 units left and 4 units up

c. Shifted 3 units right and 4 units down

d. Shifted 3 units left and 4 units down

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8 The graph of $y = f(x)$ is reflected across the x-axis and shifted down by 5 units. What is the equation of the resulting graph?

a. $y = -f(x) - 5$

b. $y = f(-x) - 5$

c. $y = -f(x - 5)$

d. $y = -f(x) + 5$

9 What is the vertex of the parabola $f(x) = -2(x + 1)^2 + 7$?

a. $(-1, 7)$

b. $(1, 7)$

c. $(-1, -7)$

d. $(1, -7)$

Section 4: Piecewise & Absolute Value Functions

10 Given the piecewise function $f(x) = 2x + 1$ for $x < 2$, and $f(x) = x^2 - 3$ for $x \geq 2$, evaluate $f(2) + f(0)$.

a. 2

b. 1

c. 3

d. 4

11 What is the range of the absolute value function $f(x) = |x - 4| + 3$?

a. $[3, \infty)$

b. $[4, \infty)$

c. $(-\infty, 3]$

d. $[0, \infty)$

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12 For what value of x does the function $f(x) = -|x + 2| + 5$ attain its maximum value?

a. -2

b. 2

c. 5

d. 0

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